

# Nadia Figueroa

Department of Mechanical Engineering and Applied Mechanics  
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🌐 <https://nbfigueroa.github.io/>

Last Updated: April 3, 2025

## EDUCATION

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- Swiss Federal Institute of Technology Lausanne (EPFL)** Oct. 2013 – Dec. 2019  
*Ph.D. Robotics, Control and Intelligent Systems* *Lausanne, Switzerland*  
Thesis: “From High-Level to Low-Level Robot Learning of Complex Tasks: Leveraging Priors, Metrics and Dynamical Systems”  
Advisor: Aude Billard  
Committee: Jamie Paik, Julie Shah (MIT), Katherine Kuchenbecker (MPI), Melanie Zeilinger (ETH)
- TU Dortmund University** Oct. 2009 – May 2012  
*M.Sc. Automation and Robotics (Robotics Major)* *Dortmund, Germany*  
\*Thesis conducted at the German Aerospace Center (DLR)
- Monterrey Institute of Technology (ITESM)** Oct. 2002 – Dec. 2007  
*B.Sc. Mechatronics (Robotics Minor)* *Monterrey, Mexico*

## ACADEMIC POSITIONS

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- University of Pennsylvania** Jan. 2022–present  
*Shalini and Rajeev Misra Presidential Assistant Professor* *Philadelphia, USA*  
Mechanical Engineering and Applied Mechanics (Primary)  
Computer and Information Science (Secondary)  
Electrical and Systems Engineering (Secondary)  
Primary Faculty of GRASP Laboratory
- Massachusetts Institute of Technology** Jan. 2020–Dec. 2021  
*Postdoctoral Associate* *Cambridge, USA*  
Computer Science and Artificial Intelligence Lab (CSAIL)  
Department of Aeronautics and Astronautics

## AWARDS AND HONORS

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- IEEE Transactions on Robotics King-Sun Fu Memorial Best Paper Award** March 2025  
\*Honorable mention for paper [J3].
- Outstanding Program Committee Member for ACM/IEEE HRI 2023** March 2023  
Nominated by the technical theme chair based on the high-quality of meta-reviews and contributions to the program committee meeting that were insightful, constructive, and thorough.
- Finalist for the 2022 Agilent Early Career Professor Award** May 2022  
For early career professors with significant original research contributions and outstanding potential for future research enabling technologies of importance to *Agilent Technologies* and the world.
- Outstanding Associate Editor for IEEE RA-L** May 2022  
The IEEE Robotics and Automation Letters (RA-L) annually recognizes Associate Editors for their distinguished service. Candidates are nominated by the Editor-in-Chief and Senior Editors.

- Finalist for the euRobotics Georges Giralt PhD Award** March 2020  
Sponsored by *euRobotics*. This award is a European scientific prize for extraordinary contributions in robotics. It is yearly awarded at the European Robotics Forum.
- Finalist for the EPFL Doctoral Program Distinction Award** June 2019  
Each year, a distinction is granted to a selection of very high quality theses, in order to highlight the doctoral candidates' research work and their scientific merit.
- Nominated for the ABB (Asea Brown Boveri) PhD Award** June 2019  
The prize is awarded, every two years, for particularly excellent doctoral work in the fields of energy and information technology and automation technology.
- Finalist for the KUKA Innovation Award** April 2017  
Sponsored by *KUKA*. This award is given to the top 5 finalists in a world-wide robotics innovation competition. Our team developed an application of multiple robot arms in collaboration with a human, spotlight video [here](#).
- Best Student Paper Award at Robotics: Science and Systems** June 2016  
Sponsored by *Springer*. Our work on coordinated multi-arm motion planning was also selected as **finalist** for Best Conference Paper Award and Best Systems Award. Spotlight video [here](#).
- Human-Robot Interaction (HRI) Pioneer** March 2016  
HRI Pioneers identifies and empowers the world's top student researchers early in their careers.

PUBLICATIONS (1159 CITATIONS, H-INDEX: 20)

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## Books

- [B1] Billard, A., Mirrazavi Salehian, S. S. and **Figueroa, N.** "Learning for Adaptive and Reactive Robot Control: A Dynamical Systems Approach." *Intelligent Robotics and Autonomous Agents Series*, MIT Press, 2022. <https://mitpress.mit.edu/9780262046169/>

## Refereed Journals and Transactions

- [J1] Sun, S., Gao, H., Li, T. and **Figueroa, N.** "Directionality-Aware Mixture Model Parallel Sampling for Efficient Linear Parameter Varying Dynamical System Learning." *IEEE Robotics and Automation Letters (RA-L)*, May 2024.
- [J2] Koptev, M., **Figueroa, N.** and Billard, A. "Reactive Collision-Free Motion Generation in Joint Space via Dynamical Systems and Sampling-Based MPC." *The International Journal of Robotics Research (IJRR)*, March 2024.
- [J3] Fourie, C., **Figueroa, N.** and Shah, J. "On-Manifold Strategies for Reactive Dynamical System Modulation with Non-Convex Obstacles." *IEEE Transactions on Robotics (TRO)*. March 2024
- [J4] Jin, L., Yang, Y., Maldonado, B.O.T., Lee, S.D., **Figueroa, N.**, Full, R.J. and Yang, S., "Ultra-fast, Programmable, and Electronics-Free Soft Robots Enabled by Snapping Metacaps." *Advanced Intelligent Systems*, Feb. 2023. <https://doi.org/10.1002/aisy.202300039>.
- [J5] Koptev, M., **Figueroa, N.**, Billard, A. "Neural Joint-Space Implicit Distance Functions for Reactive Robot Manipulator Control". *IEEE Robotics and Automation Letters (RA-L)*, vol. 8, no. 2, pp. 480-487, Feb. 2023
- [J6] **Figueroa, N.**, Billard, A. "Locally Active Globally Stable Dynamical Systems: Theory, Learning and Experiments." *The International Journal of Robotics Research (IJRR)*, 41(3):312-347, 2022
- [J7] Koptev, M., **Figueroa, N.** and Billard, A. "Real-Time Self-Collision Avoidance in Joint Space for Humanoid Robots." *IEEE Robotics and Automation Letters (RA-L)*, 6(2): 1240-1247. 2021.
- [J8] Shavit, Y., **Figueroa, N.**, et al. "Learning Augmented Joint-Space Task-Oriented Dynamical Systems: A Linear Parameter Varying and Synergetic Control Approach". *IEEE RA-L*. 2018.

- [J9] Mirrazavi Salehian, S. S., **Figueroa, N.** and Billard, A. “A Unified Framework for Coordinated Multi-Arm Motion Planning.” *The International Journal of Robotics Research*, 37(10), 1205-1232, 2018.
- [J10] **Figueroa, N.**, et al. “A Combined Approach Toward Consistent Reconstructions of Indoor Spaces Based on 6D RGB-D Odometry and KinectFusion”. *ACM TIST*, 6(2):14, 2015.
- [J11] Dong, H., **Figueroa, N.** and El Saddik, A. “Elicitation Study on Gesture Preferences and Memorability Towards a Practical Hand-Gesture Vocabulary for Smart TVs”. *IEEE Access*, 2015.
- [J12] Dong, H., **Figueroa, N.** and El Saddik, A. “Load Balance Control for a Humanoid Musculoskeletal Arm in Table Tennis”. *Intl. Journal of Control, Automation and Systems*, 4:887-896, 2015.
- [J13] Dong, H., **Figueroa, N.**, El Saddik, A. “Adaptive load-distributed muscle coordination method for kinematically redundant musculoskeletal humanoid systems”. *RAS*, 64:56-69, 2015.
- [J14] Dong, H., Ugalde, I., **Figueroa, N.** and El Saddik, A. “Towards whole body fatigue assessment of human movement”. *Sensors*, 14(2):2052-2070, 2014.
- [J15] Ali, H., Shafait, H., Giannakidou, E., Vakali, A., **Figueroa, N.**, Varvadoukas, T. and Mavridis, N. “Contextual object category recognition for RGB-D scene labeling”. *RAS*, 62(2):241-256, 2014.
- [J16] Dong, H., Giakoumidis, N., **Figueroa, N.** and Mavridis, N. “Approaching Behaviour Monitor and Vibration Indication in Developing a General Moving Object Alarm System”. *JARS*, 10, 2013.
- [J17] **Figueroa, N.**, Schmidt, F., et al. “Joint origin identification of articulated robots with marker-based multi-camera optical tracking systems”. *Robotics and Autonomous Systems*, 61(6):580-592, 2013.

#### Refereed Conference Proceedings

- [C1] Srikanthan, A.\*, Xue, Y.\*, Kumar, V., Matni, N. and **Figueroa, N.**, “ADMM-MCBF-LCA: A Layered Control Architecture for Safe Real-Time Navigation” *Accepted at at IEEE International Conferences on Robotics and Automation (ICRA) 2025, Atlanta, USA*. \*Equal contribution.
- [C2] Unger, G., Shenoy, S., Li, T., **Figueroa, N.**, and Sung, C., “MOF: Magnetic Origami Folding System for Repeatably Reconfigurable Structures with Enhanced Fold Angle Control”. *Accepted at IEEE International Conferences on Robotics and Automation (ICRA) 2025, Atlanta, USA*
- [C3] Sun, S. and **Figueroa, N.** “SE(3) Linear Parameter Varying Dynamical Systems for Globally Asymptotically Stable End-Effector Control”. *IEEE/RSJ IROS 2024, Abu Dhabi, UAE*.
- [C4] Nawaz, F., Peng, S., Lindemann, L., **Figueroa, N.** and Matni, N. “Reactive Temporal Logic-based Planning and Control for Interactive Robotic Tasks”. *IEEE/RSJ IROS 2024, Abu Dhabi, UAE*.
- [C5] Martini, E., Parekh, H., Peng, S., Bombieri, N., and **Figueroa, N.** “A Robust Filter for Marker-less Multi-person Tracking in Human-Robot Interaction Scenarios”. *IEEE RO-MAN 2024, USA*.
- [C6] Shao, Y., Li, T., Keyvanian, S., Chaudhari, P., Kumar, V., and **Figueroa, N.** “Constraint-Aware Intent Estimation for Dynamic Human-Robot Object Co-Manipulation.” *RSS 2024, Netherlands*.
- [C7] Peng, S., Wang, X. M. Wang, Shah J.A., and **Figueroa N.** “Object Permanence Filters for Robust Tracking with Interactive Robots”. *IEEE ICRA 2024, Yokohama, Japan*.
- [C8] Nawaz, F., Li, T., Matni, N. and **Figueroa, N.** “Learning Complex Motion Plans using Neural ODEs with Safety and Stability Guarantees”. *IEEE ICRA 2024, Yokohama, Japan*.
- [C9] Zhang, Z., Li, T. and **Figueroa N.** “Constrained Passive Interaction Control: Leveraging Passivity and Safety for Robot Manipulators”. *IEEE ICRA 2024, Yokohama, Japan*.
- [C10] Choi, H., and **Figueroa N.** “Towards Feasible Dynamic Grasping: Leveraging Gaussian Process Distance Fields, SE(3) Equivariance and Riemannian Mixture Models”. *IEEE ICRA 2024, Japan*.

- [C11] Choi, H., Das, S., Peng, S., Bajcsy, R. and **Figueroa N.** “On the Feasibility of EEG-based Motor Intention Detection for Real-Time Robot Assistive Control”. *IEEE ICRA 2024, Yokohama, Japan*.
- [C12] Beik-Mohammadi, H., Hauberg, S., Arvanitidis, G., **Figueroa, N.**, Neumann, G., and Rozo, L. “Neural Contractive Dynamical Systems”. *Spotlight (5% acc. rate) at ICLR 2024*.
- [C13] Keyvanian, S., Johnson, M. J. and **Figueroa, N.** “Learning Realistic Joint Space Boundaries for Range of Motion Analysis of Healthy and Impaired Human Arms.” In Proc. of the *2023 IEEE-RAS 22nd International Conference on Humanoid Robots (Humanoids)*, Austin, USA, 2023
- [C14] Li, T. and **Figueroa, N.** “Task Generalization with Stability Guarantees via Elastic Dynamical System Motion Policies”. In Proc. of the *7th Conference on Robot Learning (CoRL)*, Nov 2023
- [C15] Wang, Y., **Figueroa, N.**, Li, S., Shah, A. and Shah, J. “Temporal Logic Imitation: Learning Plan-Satisficing Motion Policies from Demonstrations.” In Proc. of the *6th Conference on Robot Learning (CoRL)*, PMLR 205:94-105, Dec 2022
- [C16] Zhou, Y., Booth, S., **Figueroa, N.** and Shah, J. “RoCUS: Robot Controller Understanding via Sampling.” In Proc. of the *5th Conference on Robot Learning*, PMLR 164:850–860, Nov 2021.
- [C17] Li, S., **Figueroa, N.**, Shah, A. and Shah, J. “Provably Safe and Efficient Motion Planning under Uncertainty for Human-Robot Collaboration.” In Proceedings of *RSS XVII*, Virtual, July 2021.
- [C18] **Figueroa, N.**, Faraji, S., et al. “A Dynamical System Approach for Adaptive Grasping, Navigation and Co-Manipulation with Humanoid Robots.” In Proc. of the *IEEE ICRA*, 2020
- [C19] **Figueroa, N.** and Billard, A. “A Physically-Consistent Bayesian Non-Parametric Mixture Model for Dynamical System Learning”. In Proc. of the *2nd CORL*. Zürich, Switzerland. October 2018.
- [C20] Mirrazavi Salehian, S. S., **Figueroa, N.** and Billard, A. “Dynamical System-based Motion Planning for Multi-Arm Systems: Reaching for Moving Objects” In Proc. of the *2017 IJCAI*. Australia.
- [C21] Mirrazavi Salehian, S. S., **Figueroa, N.** and Billard, A. “Coordinated multi-arm motion planning: Reaching for moving objects in the face of uncertainty.” In Proc. of *RSS XVI*, Michigan, USA, 2016. \**[Winner of Best Student Paper Award, Finalist for Best Conference Paper & Best Systems Paper]*
- [C22] Beetz, M., Bessler, D., Winkler, J., Bartels, G., Billard, A., **Figueroa, N.**, et al. “Open Robotics Research Using Web-based Knowledge Services.” In Proc. of the *IEEE ICRA*, Sweden, 2016.
- [C23] Dong, H., **Figueroa, N.** and El Saddik, A. “An Elicitation Study on Gesture Attitudes and Preferences Towards an Interactive Hand-Gesture Vocabulary”. In *MM '15 Proceed.*, 2015.
- [C24] Dong, H., **Figueroa, N.** and El Saddik, A. “Towards consistent reconstructions of indoor spaces based on 6D RGB-D odometry and KinectFusion”. In *IEEE IROS*, pages 1796-1803, 2014.
- [C25] Dong, H., Yazdkhasti, S., **Figueroa, N.** and El Saddik, A. “Anti-fatigue control for over-actuated bionic arm with muscle force constraints”. In *IEEE IROS*, pages 335-342, 2013.
- [C26] **Figueroa, N.**, Dong, H. and El Saddik, A. “From Sense to Print: Towards Automatic 3D Printing from 3D Sensing Devices”. In *IEEE SMC*, pages 4897-4904, 2013.
- [C27] Dong, H., **Figueroa, N.** and El Saddik, A. “Muscle Force Control of a Kinetically Redundant Bionic Arm with Real-Time Parameter Update”. In *IEEE SMC*, pages 1640-1647, 2013.
- [C28] Ali, H. and **Figueroa, N.** “Segmentation and Pose Estimation of Planar Metallic Objects”. In *Ninth Conf. on Computer and Robot Vision (CRV)*, pages 376-382, 2012.
- [C29] **Figueroa, N.**, Ali, H. and Schmidt, F. “3D registration for verification of humanoid Justin’s upper body kinematics”. In *Ninth Conference on CRV*, pages 276-283, 2012.

## Workshop Papers and Poster Presentations

- [W1] Gao, G., Li, T., and **Figueroa, N.**, “Object-Centric Out-of-Distribution Recovery with Keypoint Inverse Policy For Visuomotor Imitation Learning” *Presented at the CORL 2024 Workshop on Lifelong Learning for Home Robots*, selected for Spotlight Presentation. November 2024.
- [W2] Lee, S. and **Figueroa, N.**, “Towards Real-Time Muscle-in-the-Loop Robotic Assistants via AI-Driven Wearable Ultrasound Processing” *Presented at the 2024 Northeastern Robotics Colloquium (NERC)*, selected for Spotlight Presentation. September 2024.
- [W3] Zhang, C., Shao, Y., Parekh, H., Shi, J., Chaudhari, P., Kumar, V., and **Figueroa, N.** “Don’t Yell at Your Robot: Physical Correction as the Collaborative Interface for Language Model Powered Robots” *Presented at RSS 2024 Generative AI for HRI (genAI-HRI) Workshop*. July 2024.
- [W4] Shao, Y., Li, T., Keyvanian, S., Chaudhari, P., Kumar, V., and **Figueroa, N.** “A Dynamical System Approach to Intent Estimation and Co-Manipulation”. In Proc. of the *ICRA 2024 Workshop Towards Collaborative Partners: Design, Shared Control, and Learning for Physical Human-Robot Interaction*.
- [W5] Wang, F., **Figueroa, N.**, Li, S., Shah, A. and Shah, J. “Learning Plan-Satisficing Motion Policies from Demonstrations” In Proc. of the *IROS 2023 Workshop on Learning Meets Model-based Methods for Manipulation and Grasping*. *\*[Winner of Agile Robots Best Student Paper Award]*
- [W6] Wang, F., **Figueroa, N.**, Li, S., Shah, A. and Shah, J. “Temporal Logic Imitation: Learning LTL-Satisficing Dynamical Systems from Demonstration”. In Proc. of the *RSS 2022 Workshop on Overlooked Aspects of Imitation Learning: Systems, Data, Tasks, and Beyond*.
- [W7] Koptev, M., **Figueroa, N.**, Billard, A. “Implicit Distance Functions: Learning and Applications in Control.” In Proc. of the *2022 IEEE ICRA Workshop on Motion Planning with Implicit Neural Representations of Geometry*.
- [W8] Mirrazavi Salehian, S. S., Lin, H.C., **Figueroa, N.**, Smith, J., Mistry, M. and Billard, A. “Transitioning with confidence during contact/non-contact scenarios.” In Proc. of the *2018 IEEE IROS Workshop on Human-Robot Cooperation and Collaboration in Manipulation*.
- [W9] **Figueroa, N.** and Billard, A. “Modeling Compositions of Impedance-based Primitives via Dynamical Systems.” In Proc. of the *2018 IEEE ICRA Workshop on Cognitive Whole-Body Control for Compliant Robot Manipulation (COWB-COMP)*.
- [W10] **Figueroa, N.** and Billard, A. “Multi-Arm Self-Collision Avoidance: A Sparse Solution for a Big Data Problem.” In Proc. of the *2018 Third Machine Learning in Planning and Control of Robot Motion (MLPC) Workshop at 2018 IEEE ICRA*.
- [W11] **Figueroa, N.** and Billard, A. “Learning Complex Manipulation Tasks from Heterogeneous and Unstructured Demonstrations.” In Proc. of the *2017 IEEE/RSJ IROS Workshop on Synergies between Interaction and Learning*.
- [W12] **Figueroa, N.**, Pais, A. L. and Billard, A. “Learning Complex Sequential Tasks from Demonstration: A Pizza Dough Rolling Case Study.” In Proc. of the *2016 ACM/IEEE Intl. Conf. on Human-Robot Interaction (HRI)*. HRI 2016 Pioneers.

## Manuscripts Under Review or In Preparation

- [M1] Weakly, J., Gong, V., Kang, T., Li, T., **Figueroa, N.** and Sung, C. “KGripper: A Push-On Twist-Off Passive Perching Mechanism”. *Under Review at IEEE/ASME Transactions on Mechatronics*.
- [M2] Li, T., Sun, S., Aditya, S., and **Figueroa, N.** “Elastic Motion Policy: Real-Time Adaptive One-Shot Imitation Learning with Stability Guarantees in Full Pose”. *Under Review at IROS 2025*.

- [M3] Gao, J., Li, T. and **Figuerroa, N.** “Out-of-Distribution Recovery with Object-Centric Keypoint Inverse Policy for Visuomotor Imitation Learning”. *Under Review at IROS 2025.*
- [M4] Zhang, Z., Xue, Y., **Figuerroa, N.** and Akesson, K. “Gradient Field-Based Dynamic Window Approach for Collision Avoidance in Complex Environments”. *Under Review at IROS 2025.*
- [M5] Das, S., Xue, Y., Li, H. and **Figuerroa, N.** “RNBF: Real-Time RGB-D Based Neural Barrier Functions for Safe Robotic Navigation”. *Under Review at IROS 2025.*
- [M6] Xue, Y. and **Figuerroa, N.**, “No Minima, No Collisions: Combining Modulation and Control Barrier Function Strategies for Feasible Dynamic Collision Avoidance”. *In preparation for TRO.*
- [M7] Martini, E., Bombieri, N., and **Figuerroa, N.** “A Lightweight and Low-Bandwidth Aggregator for Multi-view 3D Human Pose Tracking”. *In preparation for Information Fusion.*

## GRANTS AND FUNDING

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- [G1] Penn Engineering AI-Enabled Systems: Safe, Explainable, and Trustworthy (**ASSET**) Center and Institute for Biomedical Informatics (**IBI**) Collaborative Research in Trustworthy AI for Medicine. *From Thought to Action: Deciphering Neuromuscular Control via Non-Invasive Biosignal Tracking for Real-Time Personalized Robotic Assistants*,  
**PI** [Co-PIs: Flavia Vitale, Ruzena Bajcsy], 05/01/2024–05/30/2025, \$100,000
- [G2] National Science Foundation - Foundational Research in Robotics (**NSF-FRR**) #2328476,  
*Travel: Inclusion at Robotics: Science and Systems 2023; Daegu, Republic of Korea; 10-14 July 2023*,  
**PI** [Co-PI: Leonardo Bobadilla (Florida International University)], 8/01/2023–7/01/2024, \$30,000.

## INVITED TALKS, SEMINARS AND LECTURES

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### **Human-Centric Robot Learning and Control for Physical Interaction: A Dynamical Systems Approach**

|                                                                                     |          |
|-------------------------------------------------------------------------------------|----------|
| Special Seminar at Amazon Robotics Research Labs. <i>Westborough, Massachusetts</i> | Jan 2025 |
| Special Seminar at TU Darmstadt. <i>Darmstadt, Germany</i>                          | Dec 2024 |

### **A Layered Control Architecture for Safe, Reactive, and Useful Physical Human-Robot Interaction**

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|----------------------------------------------------------------------------------|----------|
| Control Architectures Theory (CAT) Workshop - IEEE CDC 2024, <i>Milan, Italy</i> | Dec 2024 |
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### **From Motion to Interaction: A Dynamical Systems Approach for Human-Centric Robot Learning and Control**

|                                        |          |
|----------------------------------------|----------|
| Autonomy Talks Seminar Series, Youtube | Nov 2024 |
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### **A 2-day Workshop on “Learning for Adaptive and Reactive Robot Control”** *Imparted four 2-hour lectures across 2 days on foundational topics related to my research.*

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|--------------------------------------------------------------|----------|
| Chalmers University of Technology, <i>Göthenberg, Sweden</i> | May 2024 |
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### **Debates on the Future of Robotics Research**

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|---------------------------------------------------------------------------------------------|----------|
| <i>Debate 3: Generative AI will make a lot of traditional robotics approaches obsolete.</i> |          |
| Debated against this thesis @ ICRA 2024 <i>Yokohama, Japan</i>                              | May 2024 |

### **Safety, Adaptation and Efficient Learning in Physical Human-Robot Interaction: A Dynamical Systems Approach**

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|--------------------------------------------------------------------------------|------------|
| Chalmers University of Technology, <i>Göthenberg, Sweden</i>                   | May 2024   |
| Boston Dynamics AI Institute Robotics Seminar, <i>Cambridge, Massachusetts</i> | April 2024 |
| Oregon State University Robotics Seminar, <i>Corvallis, Oregon</i>             | Feb 2024   |

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|------------------------------------------------------------------------------------------------------------------------------|------------|
| Caltech Mechanical and Civil Engineering Department Seminar, <i>Pasadena, California</i>                                     | Feb 2024   |
| 5th Learning for Dynamics and Control Conference (L4DC), <i>Philadelphia, PA</i> [Keynote]                                   | June 2023  |
| <b>Towards Flexible Human-Centric Automation via Efficient Learning of Adaptive Skills with Control-Theoretic Guarantees</b> |            |
| IROS 2023 Workshop on Robotics & AI in Future Factory, Detroit, USA                                                          | Oct 2023   |
| <b>Collaborative Robots in the Wild: Challenges and Future Directions from a Human-Centric Perspective</b>                   |            |
| Princeton Robotics Seminar Series, Princeton University                                                                      | Dec. 2022  |
| Illinois Robotics Seminar Series, University of Illinois Urbana-Champaign                                                    | Nov. 2022  |
| Seminar at The Center for Cognition, Action, and Perception, University of Cincinnati                                        | Oct. 2022  |
| The International Symposium on Robotics Research (ISRR), Geneva, Switzerland                                                 | Sept 2022  |
| <b>Multi-arm Robots and Human Cooperation</b>                                                                                |            |
| US-EU Workshop on Intelligent Manufacturing, Prague, Czech Republic                                                          | June 2022  |
| <b>Towards Safe and Efficient Learning and Control for Physical Human Robot Interaction</b>                                  |            |
| ICLR 2022 Workshop on Generalizable Policy Learning in the Physical World.                                                   | April 2022 |
| IROS 2021 Workshop on From Human-Robot Interaction to Collaborative Control.                                                 | Sept. 2021 |
| NYU Courant Computer Science Spring 2021 Colloquium.                                                                         | April 2021 |
| University of Pennsylvania GRASP/MEAM Department Seminar.                                                                    | Mar. 2021  |
| MIT EECS Department Special Seminar.                                                                                         | Mar. 2021  |
| Yale MEMS Department Research Seminar.                                                                                       | Feb. 2021  |
| <b>Efficient Learning and Adaptive Motion Planning for and from Physical Human Robot Interaction</b>                         |            |
| MIT AeroAstro Humans interacting with Autonomy Virtual Workshop.                                                             | Jan. 2021  |
| <b>From High-Level to Low-Level Robot Learning of Complex Tasks: Leveraging Priors, Metrics and Dynamical Systems</b>        |            |
| Robotics Research Seminar, University of Washington, Seattle, USA.                                                           | Aug. 2019  |
| Robotics Research Seminar, MIT CSAIL, Cambridge, USA.                                                                        | July 2019  |
| <b>Teaching Robots Complex Manipulation Tasks from Demonstrations</b>                                                        |            |
| Swiss AI Machine Learning Meetup, Lausanne, Switzerland                                                                      | Dec 2018   |
| <b>A Physically-Consistent Bayesian Non-Parametric Mixture Model for Dynamical System Learning</b>                           |            |
| Swiss Machine Learning Day 2018, Lausanne                                                                                    | Nov 2018   |
| <b>Invariant and Weakly-Parameterized Algorithms for Efficient Robot Learning: Tackling the Micro-Data Challenge</b>         |            |
| IROS'17 Workshop on Micro-Data: The Next Frontier in Robot Learning?                                                         | Sept 2017  |
| <b>Discovering Primitive Motions from Unstructured Heterogeneous Demonstrations</b>                                          |            |
| Swiss Machine Learning Day 2014, Lausanne                                                                                    | Oct 2014   |
| <b>3D Computer Vision and Applications in Robotics and Multimedia</b>                                                        |            |
| Seminar for NYU AD Research Rounds                                                                                           | June 2013  |

## TEACHING EXPERIENCE

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|------------------------------------------------------------------|--------------------------|
| <b>University of Pennsylvania</b>                                | <i>Philadelphia, USA</i> |
| MEAM-6230: Learning and Control for Adaptive and Reactive Robots |                          |

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|------------------------------------------------------------------------------------------|------------------------------|
| <i>*New advanced graduate-level course, technical elective for robotics tracks.</i>      | Spring 2025                  |
| MEAM-3200: Introduction to Mechanical and Mechatronic Systems                            |                              |
| <i>*Undergraduate-level, compulsory for dynamics, control and robotics concentration</i> | Fall 2024                    |
| MEAM-5200: Introduction to Robotics                                                      |                              |
| <i>*Graduate-level, compulsory for robotics masters program</i>                          | Spring 2024                  |
|                                                                                          | Fall 2023                    |
|                                                                                          | Spring 2023                  |
|                                                                                          | Spring 2022                  |
| <b>École Polytechnique Fédérale de Lausanne (EPFL)</b>                                   | <i>Lausanne, Switzerland</i> |
| MICRO-570: Advanced Machine Learning (Head TA)                                           | Spring 2017                  |
|                                                                                          | Spring 2016                  |
| MICRO 401: Machine Learning Programming (Head TA)                                        | Fall 2017                    |
|                                                                                          | Fall 2016                    |
| MICRO-455: Applied Machine Learning (TA)                                                 | Fall 2015                    |
| <b>TU Dortmund University</b>                                                            | <i>Dortmund, Germany</i>     |
| Object Oriented Programming for Automation and Robotics (TA)                             | Fall 2010                    |

## STUDENT & RESEARCHER SUPERVISION

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(◇) = current, (★)=graduated, (→)=next position

### PhD Students

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|-----------------------------------------------------------------------------------|--------------|
| ◇ Ho Jin Choi (Mechanical Engineering and Applied Mechanics)                      | 2024-present |
| ◇ Farhad Nawaz (Electrical and Systems Engineering)                               | 2023-present |
| <i>*Co-advised with Nikolai Matni</i>                                             |              |
| <i>**Internship at Honda Research Institute USA during Fall 2024-Spring 2025.</i> |              |
| ◇ Yifei Simon Shao (Computer and Information Science)                             | 2023-present |
| <i>*Co-advised with Vijay Kumar and Pratik Chaudhari</i>                          |              |
| <i>**Internship at Amazon Lab126 during Fall 2024.</i>                            |              |
| ◇ Shafagh Keyvanian (Mechanical Engineering and Applied Mechanics)                | 2022-present |
| <i>*Co-advised with Michelle Johnson</i>                                          |              |
| ◇ Yifan Xue (Mechanical Engineering and Applied Mechanics)                        | 2022-present |
| ◇ Tianyu Li (Mechanical Engineering and Applied Mechanics)                        | 2022-present |
| <i>**Internship at Amazon Robotics during Fall 2024.</i>                          |              |

### Research Engineers

|                                                             |              |
|-------------------------------------------------------------|--------------|
| ◇ Sunan Sun (M.S. in Robotics, 2023)                        | 2022-present |
| ★ Harshil Parekh (M.S. in Robotics, 2023)                   | 2022-2024    |
| → Robotics Software Engineer at BotBuilt, started Fall 2024 |              |

### Visiting Students

|                                                                                |             |
|--------------------------------------------------------------------------------|-------------|
| ★ Ze Zhang (Automation, Chalmers University of Technology, Sweden)             | Fall 2024   |
| <i>*PhD Candidate and Visiting Research Scholar advised by Knut Åkesson</i>    |             |
| ★ Enrico Martini (Computer Science, University of Verona, Italy)               | Spring 2024 |
| <i>*PhD Candidate and Visiting Research Scholar advised by Nicola Bombieri</i> |             |



## Masters Theses

- ◇ Paul [Sanghyub] Lee (M.S. in Robotics) 2024-present  
Thesis: “Towards Real-Time Muscle Health Monitoring and Muscle-in-the-Loop Robotic Assistants via AI-Driven Wearable Ultrasound Processing”
- ◇ George [Jiayuan] Gao (M.S. in Robotics) 2024-present  
Thesis: “Visuomotor Object-Centric Out-Of-Distribution Recovery Policy”
- ★ Ho Jin Choi (M.S. Mechanical Engineering and Applied Mechanics) 2022-2024  
Thesis: “Gaussian Process-Based Active Exploration Strategies in Vision and Touch”  
Committee: Michael Posa and Cynthia Sung  
\*Awarded Outstanding Master’s Researcher Award in Mechanical Engineering and Applied Mechanics  
→ Summer 2024 internship at Mitsubishi Electric Research Labs (MERL) Cambridge, MA  
→ Ph.D. candidate in Mechanical Engineering and Applied Mechanics (Penn), started Fall 2024
- ★ Zhiquan Zhang (M.S. Electrical and Systems Engineering) 2022-2024  
Thesis: “Real-Time Safety-Critical Passive Torque Control with Analytic and Learning-based Constraints”  
\*Awarded Best Master’s Thesis in Electrical and Systems Engineering  
→ Ph.D. candidate at University of Illinois Urbana-Champaign (UIUC), started Fall 2024
- ★ Haihui Gao (M.S. Electrical and Systems Engineering) 2022-2024  
Thesis: “Advancing Dynamical Systems in Learning from Demonstration: A Novel Approach for Modeling Self-Crossing Trajectories” \*Dual M.S. in Mechanical Engineering and Applied Mechanics  
→ Ph.D. candidate at University of Montreal, started Spring 2025
- ★ Satya jeet Das (M.S. Electrical and Systems Engineering) 2023-2024  
Thesis: “Towards Vision-based Safety: A Real-Time Approach for Generating Barrier Functions using Depth Sensors”  
→ Ph.D. candidate at University of Southern California (USC), started Fall 2024
- ★ Haoming Li (M.S. Electrical and Systems Engineering) 2023-2024  
Thesis: “Towards Robust SLAM with Neural Implicit Representations”  
→ Reserch Engineer at the University of Pennsylvania
- ★ Dain Kim (M.Eng. in Electrical Engineering and Computer Science @ MIT) 2020-2021  
Thesis: “Imitation Learning for Sequential Manipulation Tasks: Leveraging Language and Perception”  
<https://dspace.mit.edu/handle/1721.1/139416>  
Massachusetts Institute of Technology, Co-advisor: Julie Shah (MIT)  
→ Software Engineer at Academia.edu

## Non-Thesis Masters Research

- ◇ Jalaj Shukla (M.S. in MEAM, 2025) 2025-present  
\*Independent Study Spring 2025
- ◇ Jingshuo Li (M.S. in MEAM, 2026) 2025-present  
\*Independent Study Spring 2025
- ◇ Natey Kim (M.S. in Robotics, 2026) 2025-present  
\*Independent Study Spring 2025
- ◇ Felix [Yuchen] Zheng (M.S. in Robotics, 2026) 2025-present  
\*Independent Study Spring 2025
- ◇ Zizhe Zhang (M.S. in ESE/Robotics, 2026) 2025-present  
\*Research Intern Spring 2025
- ◇ Mateusz Jaszczuk (M.S. in MEAM, 2026) 2025-present  
\*Research Intern Spring 2025

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| ◇ Ekaterina Skorniakova (M.S. in Robotics, 2026)<br>*Research Intern Fall 2024 - Spring 2025                                                                                                                 | 2024-present |
| ◇ Davin Tjandra (M.S. in MEAM, 2025)<br>*Research Intern Fall 2024 - Spring 2025                                                                                                                             | 2024-present |
| ◇ Yi-Hsuan Cheng (M.S. in Robotics, 2025)<br>*Research Intern Fall 2024, Independent Study Spring 2025                                                                                                       | 2024-present |
| ◇ Shubhodeep Shiv Aditya (M.S. in Robotics, 2025)<br>*Research Intern Summer - Fall 2024, Independent Study Spring 2025                                                                                      | 2024-present |
| ◇ Yihan Li (M.S. in Robotics, 2026)<br>*Independent Study Fall 2024                                                                                                                                          | 2024-present |
| ★ Yuxin Li (M.S. in MEAM, 2025)<br>*Independent Study Fall 2024                                                                                                                                              | 2024         |
| ★ Shaoting Peng (M.S. in Robotics, 2024)<br>*Independent Study Fall 2022, Research Intern Spring 2023-Spring 2024<br>→ Ph.D. candidate at University of Illinois Urbana-Champaign (UIUC), starting Fall 2024 | 2022-2024    |
| ★ Atharva Makarand Pradhan (M.S. in MEAM, 2024)<br>*Independent Study Fall 2023, Research Intern Spring 2024<br>→ Software Engineer at Noah Medical, started Fall 2024                                       | 2023-2024    |
| ★ Lance Tan (M.S. in DATS, 2024)<br>*Independent Study Spring 2024                                                                                                                                           | 2024         |
| ★ Parth Sanghavi (M.S. in MEAM, 2024)<br>*Independent Study Spring 2024                                                                                                                                      | 2024         |
| ★ Qi Xue (M.S. in CIS, 2023)<br>*Independent Study Fall 2023                                                                                                                                                 | 2023         |
| ★ Vaibhav Wanere (M.S. in MEAM, 2024)<br>*Independent Study Summer 2023                                                                                                                                      | 2023         |
| <b>Undergraduate Research</b>                                                                                                                                                                                |              |
| ★ Natey Kim (Bioengineering)<br>*2024 PURM Summer Scholarship Recipient                                                                                                                                      | 2024         |
| ★ Michael North (Bioengineering)<br>*2024 PURM Summer Scholarship Recipient                                                                                                                                  | 2024         |
| <b>External Collaborations</b>                                                                                                                                                                               |              |
| ★ Dr. Mikhail Koptev (Swiss Federal Institute of Technology Lausanne)<br>Advisor: Aude Billard<br>*External advisory role in thesis and collaborative publications [J2,J5,J7,W7]                             | 2019-2024    |
| ★ Dr. Christopher Fourie (Massachusetts Institute of Technology)<br>Advisor: Julie Shah<br>*Direct advisory and collaborative publication [J3]                                                               | 2021-2024    |
| ★ Dr. Hadi Beik-Mohammadi (Bosch AI/KIT, Germany)<br>Advisors: Leonel Rozo, Søren Hauberg, Gerard Neumann<br>*Collaborative publication [C12]                                                                | 2021-2024    |
| <b>Doctoral Dissertation Committees</b>                                                                                                                                                                      |              |
| ◇ Anusha Srikanthan (Electrical and Systems Engineering)<br>*Reader role. University of Pennsylvania, Advisor: Nikolai Matni and Vijay Kumar                                                                 | 2024-present |
| ◇ Sittinon ‘Non’ Nuethong (Bioengineering)<br>*Reader role. University of Pennsylvania, Advisor: Josh Baxter                                                                                                 | 2024-present |

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| ◇ Francis (Olaoluwaotan) Sowande, (Mechanical Engineering and Applied Mechanics)<br>*Reader role. University of Pennsylvania, Advisor: Michelle Johnson | 2023-present |
| ◇ Erica Waters, (Mechanical Engineering and Applied Mechanics)<br>*Chair role. University of Pennsylvania, Advisor: Michelle Johnson                    | 2023-present |
| ◇ Elizabeth “Bibit” Bianchini (Mechanical Engineering and Applied Mechanics)<br>*Chair role. University of Pennsylvania, Advisor: Michael Posa          | 2023-present |
| ◇ Tuan Anh Nguyen (Bioengineering)<br>*Reader role. University of Pennsylvania, Advisor: Michelle Johnson                                               | 2024-present |
| ◇ Victoria Edwards (Mechanical Engineering and Applied Mechanics)<br>*Chair role. University of Pennsylvania, Advisor: M. Ani Hsieh                     | 2023-present |
| ◇ Christopher Kim (Mechanical Engineering and Applied Mechanics)<br>*Reader role. University of Pennsylvania, Advisor: Cynthia Sung                     | 2023-present |
| ★ William Yang (Mechanical Engineering and Applied Mechanics)<br>*Chair role. University of Pennsylvania, Advisor: Michael Posa                         | 2023-present |
| ★ Maria-Elisabeth Tzes (Electrical and Systems Engineering)<br>*Reader role. University of Pennsylvania, Advisor: George Pappas                         | 2024         |
| ★ Christopher Fourie (Aeronautics and Astronautics)<br>*External Reader role. Massachusetts Institute of Technology, Advisor: Julie Shah                | 2024         |

## PROFESSIONAL MEMBERSHIPS

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|---------------------------------------------------|--------------|
| IEEE Member                                       | 2018-present |
| IEEE Robotics and Automation Society (RAS) Member | 2018-present |

## SERVICE

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(◇) = current role, (★)=past

### Leadership

- ◇ **Workshop Chair** for the IEEE-RAS International Conference on Humanoid Robots (HUMANOIDS 2025) at COEX, Seoul, South Korea, from September 30 to October 2, 2025
- ◇ **Industry Chair** for the IEEE/SICE International Symposium on System Integration (SII), January 2026 in Cancun, Mexico
- ◇ **Interim-Chair** of IEEE Women In Engineering (IEEE-WiE) at Robotics and Automation Society. Appointed January 2025. Have been co-chair since 2023. Spearheading the IDEA: Inclusion, Diversity, Equity & Accessibility Travel Support Program for RAS Conferences. Site: [here](#).
- ◇ **WiE Representative** of the IEEE-RAS Diversity, Equity, and Inclusion Initiatives Committee
- ★ **Bridge Programs Chair** for the 39th Annual AAAI Conference on Artificial Intelligence, February 25 - March 4, 2025 in Philadelphia, Pennsylvania, USA.
- ★ **EDI Chair** of Equity, Diversity, and Inclusion for the 2024 IEEE-RAS International Conference on Humanoid Robots (Humanoids), November 22-24, 2024 in Nancy, France.
- ★ **EDI Chair** of Equity, Diversity, and Inclusion for the 2023 IEEE-RAS International Conference on Humanoid Robots (Humanoids), December 12-13, 2023 in Austin, USA.
- ★ **Co-Chair** of the **Inclusion@RSS** Fellowship program for the 19th edition of Robotics: Science and Systems, Jul 10 – Jul 14, 2023 in Daegu, Republic of Korea. **Raised \$42k USD from NSF, Google and TRI** for an all-expenses included travel fellowship to 16 students from underrepresented background in robotics. Website: [here](#)

- ★ **Treasurer, Community Development Co-Chair** (2020 FY) and **Innovation Co-Chair** (Fall 2021) for the MIT Researchers and Postdocs Association of Aeronautics and Astronautics (RPA<sup>3</sup>).
- ★ **Postdoc representative** in the inaugural CSAIL Postdoc and Graduate Student Council (CPGSC) at MIT CSAIL led by Prof. Daniela Rus. Site: <https://www.csail.mit.edu/cpgsc>
- ★ **Postdoc representative** in the CSAIL Culture Committee led by Prof. Daniel Jackson focused on improving community and culture at CSAIL.
- ★ Part of the team of MIT postdocs that conducted and analyzed results on the “Postdoc Hiring and Recruitment at MIT Survey”. The full report can be found [here](#).
- ★ Organized and moderated the “Picture a Scientist” Virtual Film Screening and Panel Discussion in January 2021. Event details [here](#).
- ★ Mentor within the Minds Matter Boston (MMB) program for the Junior Class of 2020-2021.

#### Editorial Roles

- ◇ **Area Chair** for 2025 Robotics: Science and Systems (RSS) conference to be held June 21 – June 25, 2025 in Los Angeles, California
- ◇ **Associate Editor** for ACM Transactions in Human-Robot Interaction (THRI), start October 2024.
- ◇ **Associate Editor** for the “Robot Learning 2” Area of the IEEE Robotics and Automation Letters (RA-L), appointed July 2021 - to date, began second 3-year term in July 2024.
- ◇ **Area Chair** for the 7th Annual Learning for Dynamics & Control Conference (L4DC) to be held in the University of Michigan, Ann Arbor on June 4-6, 2025.
- ★ **Associate Editor** for the 2025 IEEE International Conference on Robotics & Automation (ICRA) to be held on 19–23 May in Atlanta, USA.
- ★ **Area Chair** of Systems Theme for the 2025 ACM/IEEE International Conference on Human Robot Interaction (HRI) to be held from March 4-6, 2025 in Melbourne, Australia
- ★ **Associate Editor** for International Symposium on Robotics Research (ISRR) 2024 to be held from December 8-12 in Long Beach, California.
- ★ **Area Chair** for the 5th Annual Learning for Dynamics & Control Conference (L4DC). University of Pennsylvania June 14-16, 2023 in Philadelphia, Pennsylvania.
- ★ **Area Chair** of Technical Theme for the 2023 ACM/IEEE International Conference on Human Robot Interaction (HRI) held from March 13-16, 2023 in Stockholm, Sweden.
- ★ **Associate Editor** for the The International Symposium on Robotics Research (ISRR) 2022 held from September 26-20 in Geneva, Switzerland.

#### Workshop & Tutorial Organization

- ◇ **Co-Organizer** Workshop on “Structured Learning for Efficient, Reliable, and Transparent Robots” at *International Conference on Robotics and Automation (ICRA)* to be held in Atlanta, USA May 2025.
- ★ **Co-Organizer** Workshop on “Generative Modeling meets HRI (GenAI-HRI)” at 2024 *Robotics Science and Systems (RSS)* conference. July 15, 2024. Delft, Netherlands, Website [here](#)
- ★ **Co-Organizer** Workshop on “Geometry, Physics, and Human Knowledge as Inductive Bias in Robot Learning” at the 2022 *Conference on Robot Learning (CORL)*. Auckland, New Zealand. Website [here](#)
- ★ **Co-Organizer** “2nd Workshop on Bi-manual Manipulation” at the 2022 *IEEE International Conference on Robotics and Automation (ICRA)*. Philadelphia, USA. Website [here](#)
- ★ **Co-Organizer** “Workshop on Collaborative Robots and the Work of the Future” at the 2022 *IEEE International Conference on Robotics and Automation (ICRA)*. Philadelphia, USA. Website [here](#)

- ★ **Co-Chair** of the IEEE RAS Technical Committee on Collaborative Automation for Flexible Manufacturing (since August 2021).
- ★ **Co-Organizer** “Workshop on Joint Action, Adaptation and Entrainment” at the Virtual 2022 *ACM/IEEE International Conference on Human-Robot Interaction (HRI)*. Website [here](#).
- ★ **Co-Organizer** “Tutorial on Dynamical System-based Learning from Demonstration” at the *2019 International Conference on Robotics and Automation (ICRA)* in Montreal, Canada. Website [here](#).
- ★ **Co-Organizer** “Tutorial on Dynamical System-based Learning from Demonstration” offered at the *2018 Robotics: Science and Systems Conference (RSS) conference*. [Tutorial cancelled due to US immigration policy. Materials presented at EPFL and available [here](#)]

#### Journal Reviews

- International Journal of Robotics Research
- IEEE Transactions on Robotics
- IEEE Robotics and Automation Letters
- IEEE Robotics and Automation Magazine
- ACM Transactions in Human-Robot Interaction
- IEEE Transactions on Cognitive and Developmental Systems
- International Journal of Social Robotics

#### Conference Reviews

- IEEE/RSJ International Conference on Intelligent Robots and Systems
- IEEE International Conference on Robotics and Automation
- Robotics: Science and Systems
- Conference on Robot Learning
- IEEE-RAS International Conference on Humanoid Robots

#### Grant Proposal Reviews

- National Science Foundation (FRR)

### DEPARTMENTAL SERVICE

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|-----------------------------------------------------------------------|------------------|
| MEAM PhD Admissions Committee                                         | 2022, 2023, 2024 |
| Member of 6 MEAM PhD Dissertation Committees (Reader and Chair roles) | 2023-present     |
| Member of 2 Bioengineering PhD Dissertation Committees (Reader roles) | 2024-present     |
| Member of 2 ESE PhD Dissertation Committee (Reader role)              | 2024             |
| Member of 5 MEAM PhD Qualifying Exam Committees (Math Examiner)       | 2022, 2024       |
| Member of 1 CIS WPE-II Exam Committee                                 | 2022             |

### PRESS COVERAGE

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- Featured quotes in “*MIT Engineers Aim to Advance Household Robots With AI Integration*” by PYM-NTS. April 8, 2024.
- Featured quotes in “*What are industrial robots?*” by Brooke Becher. BuiltIn. November 13, 2023.
- “*Getting dressed with the help from robots,*” by Rachel Gordon. MIT News. July 12, 2021.  
Mentioned in IEEE Spectrum, TechCrunch, FastCompany, The Next Byte Podcast, etc.
- “*More transparency and understanding into machine behaviors,*” by Rachel Gordon. CSAIL and MIT News. March 22, 2021.